Practice assignment 9 - Not Graded

This assignment will not be graded and is only for practice.

Multiple Select Question (MSQ)

1 point

Consider a function $f: R^2 \rightarrow R$f : R2 → R defined as:

$$f(x, y) = \begin{cases} \frac{2xy}{\sqrt{2(x^2+y^2)}} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0). \end{cases} \text{f(x, y) = \{ 2xy1if (x, y) = (0, 0)if (x, y) = (0, 0).}$$

Choose the set of correct options.

- [ ] $|x| \leq \sqrt{x^2+y^2}$ |x| ≤ x2 + y2 and $|y| \leq \sqrt{x^2+y^2}$ |y| ≤ x2 + y2 .
- [ ] If $(x, y) \rightarrow (0, 0)$(x, y) → (0, 0), then $f(x, y) \rightarrow 0$f(x, y) → 0.
- [ ] $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$(x,y)→(0,0)lim f(x, y) exists.
- [ ] $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$(x,y)→(0,0)lim f(x, y) = 0.
- [ ] $f(x, y)$f(x, y) is continuous at the origin (0,0).

No, the answer is incorrect.

Score: 0

Accepted Answers:

$|x| \leq \sqrt{x^2+y^2}$ |x| ≤ x2 + y2 and $|y| \leq \sqrt{x^2+y^2}$ |y| ≤ x2 + y2 .

If $(x, y) \rightarrow (0, 0)$(x, y) → (0, 0), then $f(x, y) \rightarrow 0$f(x, y) → 0.

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$(x,y)→(0,0)lim f(x, y) exists.

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$(x,y)→(0,0)lim f(x, y) = 0.

1 point

Consider the following functions

$$f(x, y) = (\sqrt{1-x}, \sqrt{y})\text{f(x, y) = (1 − x}$$

, y
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)

$$g(x, y) = \sin((x^2 - 1)^2 + y^4)g(x, y) = \sin((x2 - 1)2 + y4)$$

$$h(x, y) = \frac{x}{x^2 + y^2}h(x, y) = x2+y2x$$

$$s(x, y) = x^2 e^{y^2}\ s(x, y) = x2ey2$$

Which of the following is true?

☐ Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}f = \{(x, y) \mid x \leq 1, y \geq 0\}$

☐ Domain of $h = R^2$ h = R2

☐ $g \circ f(x, y) = \sin(x^2 - 1 + y^2)g \circ f(x, y) = \sin(x2 - 1 + y2)$

☐ $((g \circ f) \times h)(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}((g \circ f) \times h)(x, y) = x2+y2\sin(x2+y2)$

☐ $\lim_{(x, y) \to (0, 0)}((g \circ f) \times h + s)(x, y) = 0(x,y)\to(0,0)\lim((g \circ f) \times h + s)(x, y) = 0$

No, the answer is incorrect.
Score: 0
Accepted Answers:
Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}f = \{(x, y) \mid x \leq 1, y \geq 0\}$
$\lim_{(x, y) \to (0, 0)}((g \circ f) \times h + s)(x, y) = 0(x,y)\to(0,0)\lim((g \circ f) \times h + s)(x, y) = 0$

1 point

Match the functions of two variables in Column A with their graphs given in Column B.

☐ i → → 4, ii → → 3

☐ i → → 3, ii → → 4

☐ iii → → 1, iv → → 2

☐ iii → → 2, iv → → 1

No, the answer is incorrect.
Score: 0
Accepted Answers:
i → → 3, ii → → 4
iii → → 1, iv → → 2

1 point

Consider two functions $f: R^2 \to R^2$ and $g: R^2 \to R$f : R2 → R2 and g : R2 → R defined as:

$$f(x, y) = (x^2, y^2)f(x, y) = (x2, y2)$$

and

$g(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$ g(x, y) = {x2+y2x2−y20if (x, y) = (0, 0)if (x, y) = (0, 0)

Which of the following option(s) is (are) true?

- [ ] $g \circ f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$ g ∘ f(x, y) = {x2+y2x2−y20if (x, y) = (0, 0)if (x, y) = (0, 0)
- [ ] $g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$ g ∘ f(x, y) = {x4+y4x4−y40if (x, y) = (0, 0)if (x, y) = (0, 0)
- [ ] $\lim_{(x, y) \to (0, 0)} g(x, y)$ (x,y)→(0,0)lim g(x, y) exists.
- [ ] $\lim_{(x, y) \to (0, 0)} g \circ f(x, y)$ (x,y)→(0,0)lim g ∘ f(x, y) exists.
- [ ] $\lim_{x \to 0} \lim_{y \to 0} g \circ f(x, y) = \lim_{y \to 0} \lim_{x \to 0} g \circ f(x, y)$ x→0lim y→0lim g ∘ f(x, y) = y→0lim x→0lim g ∘ f(x, y)
- [ ] $\lim_{x \to 0} \lim_{y \to 0} g(x, y) = 1$ x→0lim y→0lim g(x, y) = 1
- [ ] $\lim_{y \to 0} \lim_{x \to 0} g \circ f(x, y) = -1$ y→0lim x→0lim g ∘ f(x, y) = −1
- [ ] $\lim_{(x, y) \to (1, 1)} g(x, y)$ (x,y)→(1,1)lim g(x, y) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$ g ∘ f(x, y) = {x4+y4x4−y40if (x, y) = (0, 0)if (x, y) = (0, 0)
$\lim_{x \to 0} \lim_{y \to 0} g(x, y) = 1$ x→0lim y→0lim g(x, y) = 1
$\lim_{y \to 0} \lim_{x \to 0} g \circ f(x, y) = -1$ y→0lim x→0lim g ∘ f(x, y) = −1

Numerical Answer Type (NAT)
Consider a function $f(x, y) = xy$ f(x, y) = xy. If the directional derivative of f f at point (a, b)(a, b), in direction of (1,-1) and (1,1) are 1 and 5 respectively. Then find the value $\frac{2a + b}{\sqrt{2}}$ 2a+b2

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 7
1 point
Consider the following statements:

Statement 1: Let $f: R^2 \to R$f : R2 → R be a function such that $\lim_{x \to a} \lim_{y \to b} f$x → alim y → blim f exists, where $a, b \in R$a, b ∈ R. Then $\lim_{(x, y) \to (a, b)} f$(x,y) → (a,b)lim f exists.

Statement 2: Let $f: R^2 \to R$f : R2 → R be a function such that $\lim_{y \to b} \lim_{x \to a} f = f(a, b)$y → blim x → alim f = f(a, b), where $a, b \in R$a, b ∈ R. Then ff is continuous.

Statement 3: Let $f: R^2 \to R$f : R2 → R be a function such that $\lim_{x \to a} \lim_{y \to b} f = \lim_{y \to b} \lim_{x \to a} f = f(a, b)$ x → alim y → blim f = y → blim x → alim f = f(a, b), where $a, b \in R$a, b ∈ R. Then ff is continuous.

Statement 4: Let $f: R^2 \to R$f : R2 → R be a function such that $f_x(a, b)$fx(a, b) exists. Then ff is continuous.

Statement 5: Let $f: R^2 \to R$f : R2 → R be a function such that $f_y(a, b)$fy(a, b) exists. Then ff is continuous.

How many statements are correct?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 0
1 point

Consider the following two functions $f_1, f_2: R^2 \to R$f1, f2 : R2 → R defined as:

$$f_1(x, y) = xy + 2x\text{f1(x, y) = xy + 2x}$$
$$f_2(x, y) = x^2 + 2y + 3x^2 y\text{.f2(x, y) = x2 + 2y + 3x2y.}$$

Define a matrix AA as follows:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x}(3, 0) & \frac{\partial f_2}{\partial x}(1, 1) \\ \frac{\partial f_1}{\partial y}(3, 2) & \frac{\partial f_2}{\partial y}(\sqrt{7}, 2) \end{bmatrix}\text{.A = [∂x∂f1(3, 0)∂y∂f1(3, 2)∂x∂f2(1, 1)∂y∂f2 (37 }$$

$$\sqrt{\qquad\qquad\qquad

Use this information to answer questions 8,9, and 10.

1 point

Which of the following option(s) is (are) true?

☐ $\lim_{(x,\,y)\,\to\,(1,\,1)} T(x,\,y) = \lim_{x\,\to\,1}\lim_{y\,\to\,1} T(x,\,y)$(x,y)→(1,1)lim T (x, y) = x→1lim y→1lim T (x, y)

☐ $T(x,\,y)$T (x, y) is continuous at (1,1)

☐ $T(x,\,y)$T (x, y) is a linear function.

☐ $T(c\,x, c\,y) = c\;T(x,\,y)$T (cx, cy) = c T (x, y).

☐ If the earthworm approaches the point $(1,\,2)(1, 2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,\,y)\,\to\,(1,\,1)} T(x,\,y) = \lim_{x\,\to\,1}\lim_{y\,\to\,1} T(x,\,y)$(x,y)→(1,1)lim T (x, y) = x→1lim y→1lim T (x, y)

$T(x,\,y)$T (x, y) is continuous at (1,1)

If the earthworm approaches the point $(1,\,2)(1, 2)$ from any direction, then the body temperature of the earthworm approaches 12 units.

If the gradient of $T(x,\,y)$T (x, y) is (25, 18) at a point, then find the body temperature of the earthworm at that point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 77

1 point

1 point

Which of the following option(s) is (are) true?

☐ If the earthworm moves parallel to the X − X−axis, then rate of change of its body temperature at the point $(a,\,b)$(a, b) is given by $4a$ + $3b$4a + 3b.

☐ If the earthworm moves parallel to the Y − Y −axis, then rate of change of its body temperature at the point $(a,\,b)$(a, b) is given by $2a$ + $3b$2a + 3b.

☐ If the earthworm moves parallel to the straight line joining the origin and the point $(2,\,3)(2, 3)$, then rate of change of its body temperature at the point $(1,\,1)(1, 1)$ is given by $\frac{29}{\sqrt{13}}$ $\sqrt{13}$ 29.

☐ If the earthworm moves parallel to the straight line joining the origin and the point $(0,\,1)(0, 1)$, then rate of change of its body temperature at the point $(1,\,2)(1, 2)$ is given by 1010.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the earthworm moves parallel to the X − X−axis, then rate of change of its body temperature at the

point (a, b)(a, b) is given by $4a$ + $3b$4a + 3b.
If the earthworm moves parallel to the straight line joining the origin and the point $(2, 3)$(2, 3), then rate of change of its body temperature at the point $(1, 1)$(1, 1) is given by $\frac{29}{\sqrt{13}}$ 13

$\sqrt{}$
29.

Check Answers